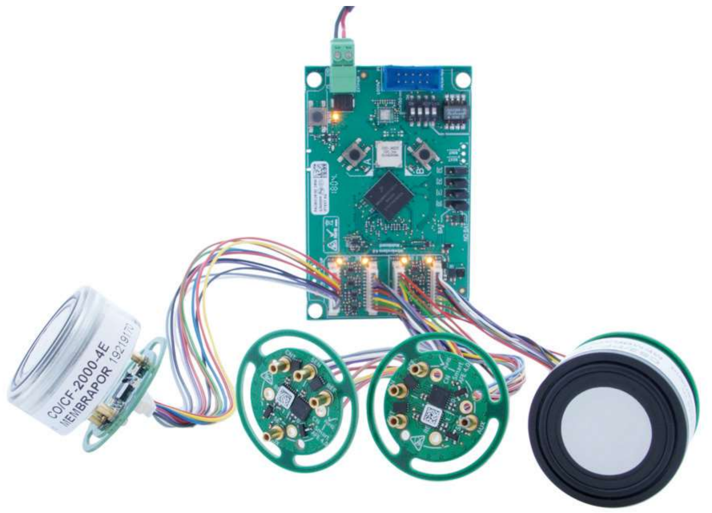
MEMBRAPOR

ELECTROCHEMICAL GAS SENSORS

MembraSens 4

Handbook



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1 General Information

1.1 What is *MembraSens*?

MembraSens is an electronic platform for the operation of electrochemical gas sensors. This can be set up with little effort, a high-quality monitoring system or a measuring instrument for the detection of toxic gases.

MembraSens consists of a motherboard with microcontroller, MODBUS interface and up to four connected analog front-ends. Such an analog front-end (AFE) includes a potentiostat for operating an electrochemical gas sensor and an EEPROM memory for securing sensor-specific data such as calibration parameters. All common gas sensors with 3 or 4 electrodes can be used with *MembraSens*.

The software on the microcontroller controls all hardware functions needed for measurements and calibrations.

1.1.1 Hardware skills

The electronic platform *MembraSens* has the following features:

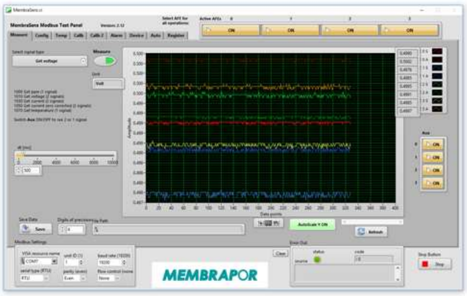
- Suitable for use with up to 4 electrochemical sensors in the Compact and Prime housing (7 series)
- For 3-electrode sensors (1 signal)
- For 4-electrode sensors (2 signals)
- ARM Cortex M4 microcontroller
- RS-485 compatible (TIA / EIA-485)
- Supply: 4 ... 36 VDC
- Defined states for sensors in de-energized state: shorted, open, maintaining bias with backup battery
- Zero point calibration possible directly with a button on the main board
- Buzzer for alerting
- Temperature measurement at the AFE directly below the gas sensor
- Two EEPROM-memories on the AFE to distinguish from factory presets
- Minimized size: AFE diameter only 24 mm, main board approx. 60x 92 mm2

1.1.2 Software skills

The sensor electronics with their firmware have the following features:

- MODBUS RTU: Digital communication for configuration, calibration and signal output of all AFEs
- Bus operation with up to 247 *MembraSens* possible
- Programmable SmartAFE: Full range of electrochemical toxic gas sensors covered
- Calibration with and without gas can be carried out arbitrarily
- Sensor signals as currents, voltages and as ppm-display available
- Temperature compensation of the baseline and the sensitivity of sensors
- Backup of the configuration, calibration and temperature compensation on the AFE in 2 different EEPROM memory (factory and user side)
- Extensive signal processing possible:
  - Various models for ppm calculation, incl. Hydrogen compensation
  - Various models of temperature compensation
- Output of the temperature
- Backup of the configuration and calibration
- Alerting when alarm thresholds are exceeded
- Bootloader for firmware update

1.1.3 Application



MembraSens can be integrated into an existing MODBUS network or connected to a PC. MEMBRAPOR's *MembraSens Modbus Test Panel* program makes it easy to configure, calibrate and measure with *MembraSens*.

1.2 The Main Board

The main board includes an ARM Cortex M4 micro controller (MC), which controls all the functionality of *MembraSens* and performs all necessary calculations. The MC collects the signals from the AFEs and executes the commands sent via MODBUS.

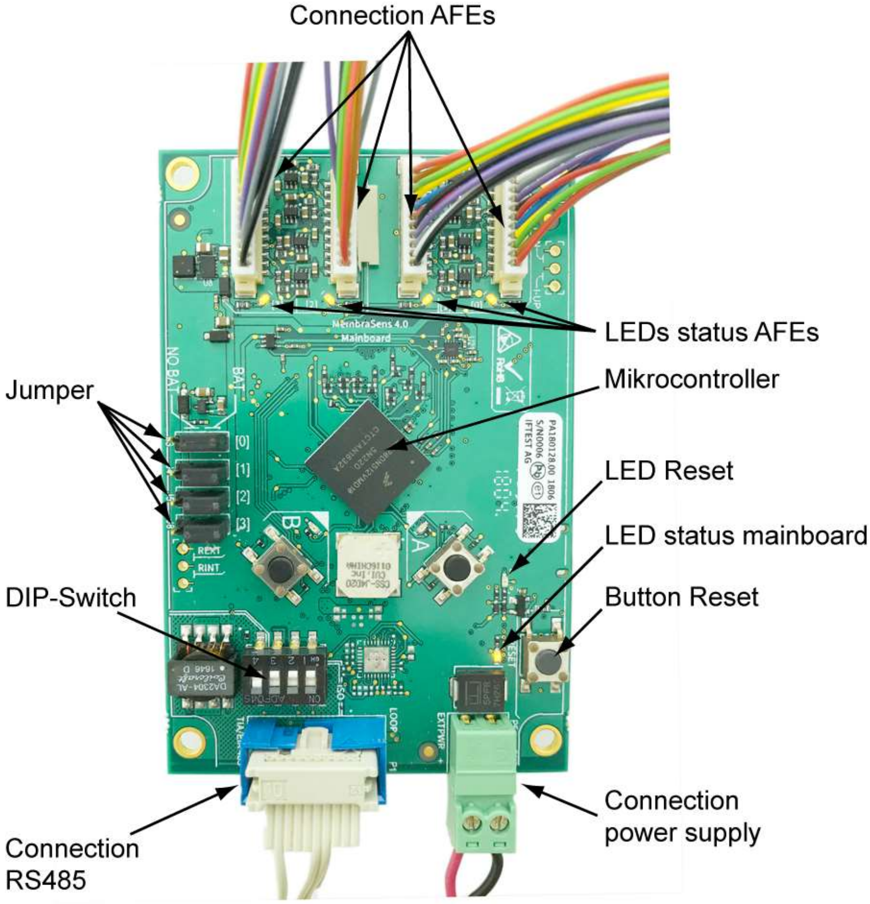


Fig. 1: The main board of MembraSens

1.3 The Smart Analog Front End (AFE)

1.3.1 Function of the AFE

The gas sensor is placed on the analogue front-end (AFE). All electrochemical gas sensors in the Compact and Prime housing (7-Series) can be used: www.membrapor.ch/compact

Fig. 2: The AFE with 4 sockets for inserting a gas sensor. The socket diameter is designed for 1.0 mm pin plug.

The smart AFE contains a potentiostatic circuit to operate an electrochemical sensor. It provides two analogue voltage signals and therefore also allows the use of 4-electrode sensors. The voltage signals are proportional to the generated sensor current and the relevant parameters of the potentiostatic circuit are programmable. With the integrated temperature sensor, the temperature is measured as close as possible to the gas sensor.

The smart AFE contains an EEPROM-memory in which the user can save the configuration, calibration parameters, temperature compensation and other data. The AFE also contains a second EEPROM-memory in which the data can be saved as a factory setting.

1.3.2 Electrochemical gas sensors

ATTENTION

It is strongly recommended to use MEMBRAPOR gas sensors to ensure the full functionality of *MembraSens*.

Gas sensors with the following electrical property can be used with the AFE:

Description	Min	Max	Unit
Sensitivity of the gas sensor	0.5	9000	nA / ppm
Sensor current	0.004	750	uA

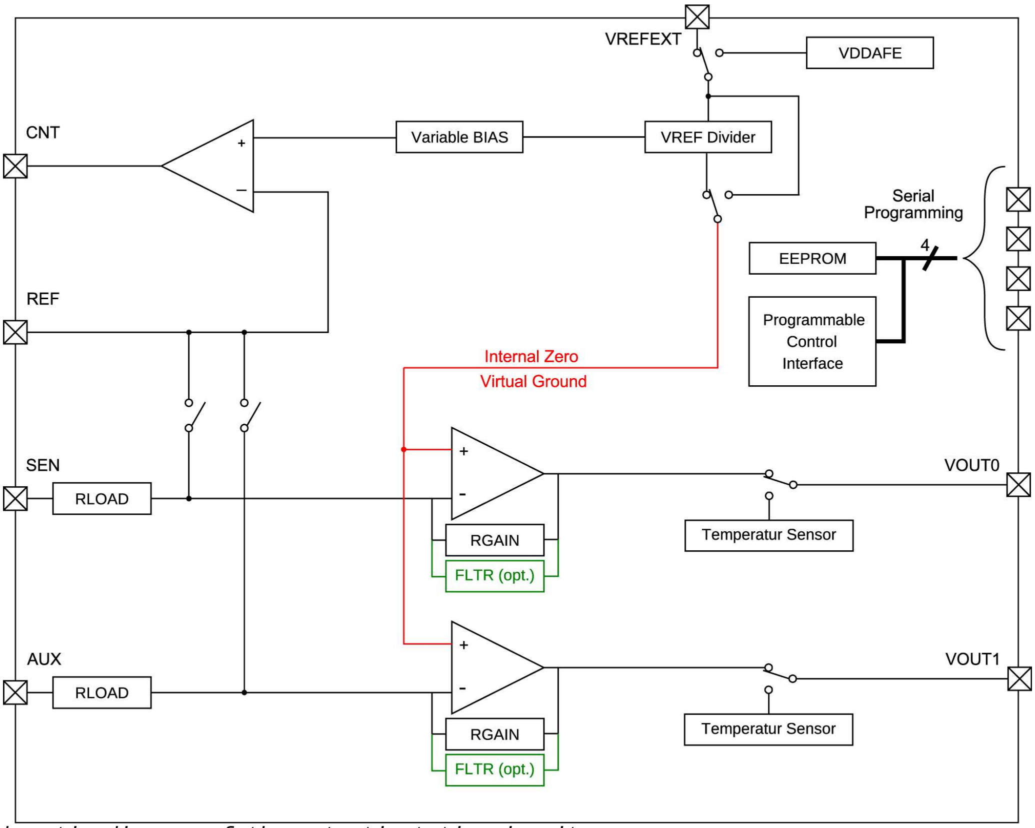
Tab 1: Specification of supported gas sensors

Sensors can be used for both reducing and oxidizing gases. For sensors that require a bias voltage, values between -600 mV to + 600 mV can be set.

Both 3-electrode and 4-electrode sensors can be used. In the former, the second signal is irrelevant.

A selection of MEMBRAPOR gas sensors can be found at:
www.membrapor.ch/compact_en

1.3.3 The potentiostatic circuit



`Fig. 3: Schematic diagram of the potentiostatic circuit`

Two potentiostatic amplifiers compare the voltages at the sensing electrode (SEN) and the auxiliary electrode (AUX) with the voltage at the reference electrode (REF) to set the bias required for operation. An occurring voltage difference is passed on to the counter electrode (CNT) for compensation.

The two transimpedance amplifiers supply voltages which are proportional to the currents through the measuring and auxiliary electrodes. These two output voltages are referenced to the internal zero point (virtual ground).

Instead of the sensor signal the voltage signal from the temperature sensor can be obtained by switching.

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2 Ports and Interfaces

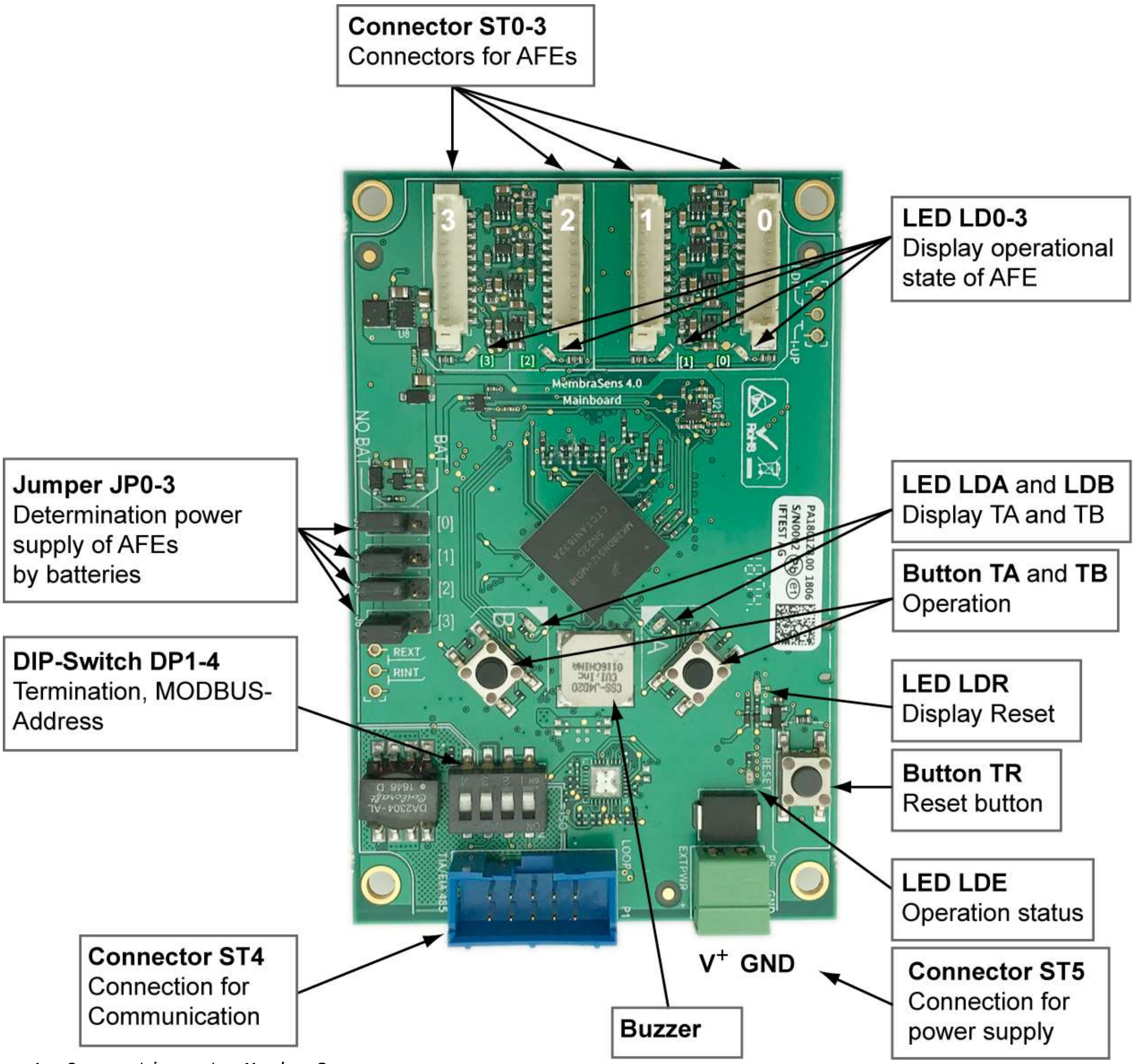

Fig. 4: Connections to MembraSens

2.1 Connection Supply Voltage

MembraSens has been designed for operation with an input voltage range of 4 to 36VDC. The supply voltage must be connected via plug ST5.

Description	**Explanation**
V^+ (EXTPWR)	Supply voltage 4 up to 36VDC
GND	Mass for V^+

Tab 2: Required supply voltage

2.2 Digital Interface for MODBUS

MembraSens has an RS485 interface in accordance with the TIA / EIA-485 standard. The MODBUS connection is made via this serial interface in 2-wire operation. *MembraSens* is to be regarded as slave in the MODBUS master-slave protocol. Operation takes place in unicast mode: The master (for example a PC) addresses a slave (an individual device, such as *MembraSens*), which processes the message after receiving it and generates a response.

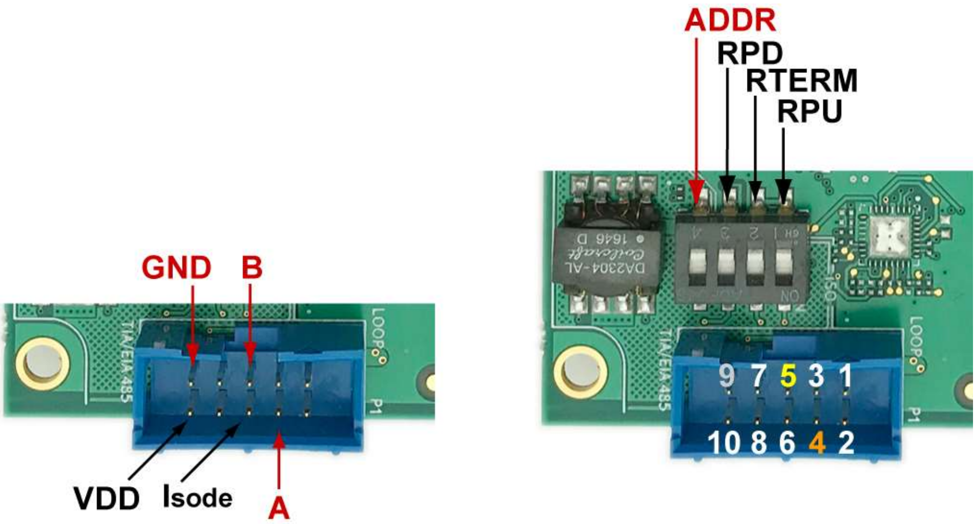

Fig 5: Plug ST4 for the MODBUS interface and DIP switch DP1-4 for termination and MODBUS address.

All communication takes place via this interface, as well as the transmission of the sensor signals.

Description	Explanation	Pin
A (orange)	Non-inverted data line	4
B (yellow)	Inverted data line	5
GND (gray)	Common ground in the bus system	9

Tab 3: Connection of the MODBUS interface

2.2.1 MODBUS RTU transmission mode

The data transfer with *MembraSens* takes place via MODBUS RTU. It must be done with 8 data bits in any case.

Parameters	Value
Baud-Rate	19,200
Data bits	8
Start bit	1
Parity	Even
Stopbit	1
Word order	CDAB
Flow control	none

Tab 4: Parameters for the MODBUS RTU data transmission

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2.2.2 MODBUS device address

By default the dip-switch DP4 is set to OFF and the device address of *MembraSens* is 1. *MembraSens* can be assigned a device address (MODBUS address) between 2 to 247, which is active when DP4 is set to ON.

To change the device address, proceed as follows:

1. Established MODBUS connection with device address 1 and DP4 to OFF
2. Reads the value (UINT16) from register 1110. If the value 1 comes back, then the data transfer works
3. Writes in register 1110 the desired device address (UNIT16)
4. Writes in register 1111 the value 0 (UNIT16) to keep the transmission mode RTU
5. Changes position of DP4 from OFF to ON
6. Performs reset by pressing the TR button

After reboot, *MembraSens* is available via the selected device address in the MODBUS network, if DP4 is ON.

2.2.3 Termination in the MODBUS network

Resistors can be switched off with DP1-3 if it is necessary in the bus system.

DP	Abbreviation	Description	Default value
1	RPU	Fail-safepull-up resistor (390 Ω)	ON
2	RTERM	Parallel termination resistor (220 Ω) between lines A and B	ON
3	RPD	Fail-safe pull-down resistor (390 Ω) on inverting line B	ON
4	ADDR	Software-controlled device address	OFF

Tab 5: Functions of dip switches DP1 to DP4

2.3 Status-LEDs

If there is a power supply to *MembraSens*, the LED LDE lights to indicate that the motherboard is ready.
The LEDs LD0 - LD3 light up depending on whether an AFE is plugged in there.
More information in chapter 5.

3 Commissioning

3.1 Power Off Condition

If *MembraSens* is not continuously connected to an external power supply in the application, but after switching on the sensors should be ready for use immediately, a corresponding state for each sensor can be set individually. These power-off states are set with jumpers JP0 - JP3 on the motherboard and resistors J8 and J9 on the AFEs.

Basically, two possible states are distinguished without external power supply. On the one hand, a further operation with button cell batteries, in which the potentiostat is operated on the AFE and the sensor is ready for use immediately after switching on the external power supply. On the other hand parking the sensors without button cell batteries, in which the potentiostat is not operated.

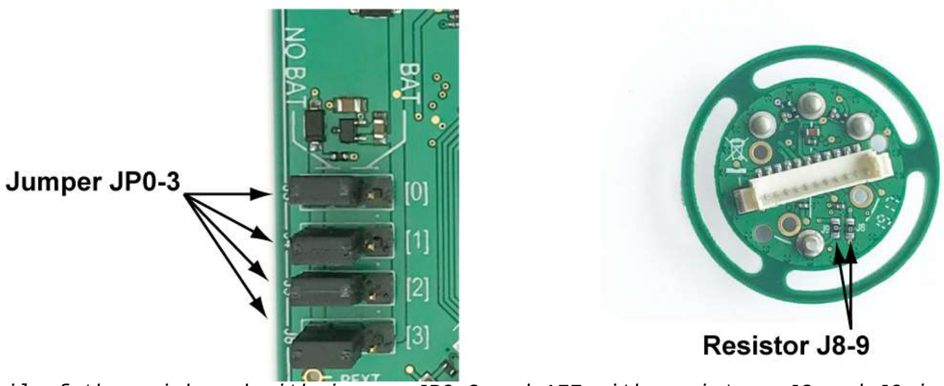

Fig.6: Detail of the mainboard with jumper JP0-3 and AFE with resistors J8 and J9 in the standard configuration.

3.1.1 Sensor continued operation without external power supply

The sensors should continue to be operated, even if the external power supply is switched off. As a result, they are ready for use immediately after switching on the device.

Configuration

- Button cell batteries must be inserted
- Jumpers must be set **towards the center** of the board

Sequence

A possible bias voltage required by the sensor is maintained. The sensors are in operation and keep their baseline. If the device is switched on, then the sensors are immediately ready for operation.

Variant

Sensors that do not require bias voltage can be shorted. To do this, set the jumper in the direction of the board edge. So the sensor does not draw power from the battery. After switching on, however, it takes up to 3 minutes for the sensor to be ready for use.

3.1.2 Parking the sensors without external power supply

Sensors should be parked if *MembraSens* is without external power supply. The use of button cell batteries is omitted.

Configuration

- No button cell batteries used
- Jumpers must be set towards the edge of the board
- For AFEs with bias voltage, the two resistors J8 and J9 must be removed

Sequence

Sensors without bias voltage are short-circuited. When the device is switched on, the sensors are stabilized within 3 minutes.

Sensors with bias voltage are not short-circuited. After switching on the device, the initially high sensor signal sinks in the direction of its baseline. Depending on the duration of the previous OFF time, this may take several hours.

3.2 Connecting *MembraSens*

The supply voltage is connected via ST5, the connection to the MODBUS via ST4. If you have questions about the right connectors, contact Membrapor.

3.3 Configure the AFEs

The AFEs are already factory configured. The internal zero line is set to 0.50V and the TIA gain to 14K ohms. This typical configuration is suitable for sensors with a medium sensitivity and positive signal (oxidation on measuring electrode).

When configuring, make sure that the amplified sensor signal always remains in the range of 0.0 - 3.0V . Signals above 3.2 V are in overload.

4 Operation

4.1 Operation via Push Buttons and the Meaning of the LEDs

LED LD0-3 (orange) lights up constantly:

Connected AFE detected.

LED LD0-3 (orange) flashes:

Warns that the sensor has exceeded the set alarm threshold at the corresponding position.

Button TA:

To acknowledge an alarm. The buzzer then switches off.

Button TR:

With the reset button the system can be restarted at any time. By pressing, the LED LDR lights up, then all LEDs flash during startup.

Perform zero calibration with buttons

MembraSens offers the possibility of performing a zero calibration of one or more gas sensors at any time directly on the board.

Output situtation:
MembraSens is *connected* to power supply.
The yellow LED lights up when connecting the power supply.
The yellow LEDs on the connections of the AFEs, in which an AFE is connected, light up.

1. Selection of the AFE with which the zero point calibration is to be carried out
<u>Press button A (for mind. 5 s)</u>
-> Red LED on button A lights up.
- > Yellow LED on AFE0 lights up, if an AFE is connected there.
Otherwise, the yellow LED on the next connection with AFE will light up.
All other yellow LEDs on the other AFEs are not lit (anymore).

Cancellation: Anytime possible with Reset Button

Next: Select AFE with button B
<u>Press button B</u>
-> LED at AFE0 goes out, LED at AFE1 lights up (if AFE is connected there, otherwise at next one)
Press button B again
-> LED at AFE1 goes out, LED at AFE2 lights up (if AFE is connected there, otherwise at next one)
Press button B again
-> LED at AFE2 goes out, LED at AFE3 lights up (if AFE is connected there, otherwise at next one)>
Press button B again
-> All LEDs on the AFEs, which areconnected, light up
Press button B again
-> LED on AFE0 lights up (if AFE is connected there), all other LEDs on the other AFEs are not lit.>
Press button B again
-> LED at AFE0 goes out, LED at AFE1 lights up (if AFE is connected there, otherwise at next one)
etc...

Intermediate Result: One or more AFEs are selected.

2. Perform zero calibration (with one or more AFEs)
<u>Press button A</u>
-> Execution of the command "Start Calibration mode" with selected AFE>
-> Yellow, lit LED on selected AFE starts flashing

After a waiting period of 3 s
-> LED on selected AFE stops flashing and only lights up.
-> Execution of the command "Calibrate zero point" with selected AFE
-> Acoustic sound

After a waiting period of 2 s
-> Execution of the "End calibration mode" command with selected AFE>
-> 2x Acoustic sound
-> Red LED on button A goes out

3. Condition again as in the initial situation
-> The yellow LEDs light up at the connections of the AFEs, where an AFE is also connected.

Keycombination for bootloader

The following key configuration is reserved for importing a new firmware: Simultaneous pressing of the keys A, B and Reset.

4.2 Operation with MODBUS Commands

In this chapter the indexes are listed thematically for the MODBUS control commands.

- The registers are addressed with 16 bits and Word-order CDAB.
- The registers have the following access rights: Read and Write. For some registers both can be done, for others only one option.
- There are registers through which a process is *initiated* at *MembraSens*. This is done as a write command with "0" as data content. Examples: Save configuration, Start calibration mode, Calibrate zero point, etc.

4.2.1 Abbreviations

Abbreviations	Meaning
Reg.-Adr.	Register address (start address)
W/R	Writing / reading
Data Contents	Data content description
AW	Number of values
Data type	Data type
AR	Number of U16 registers
A	Addresses the command an AFE (1 = yes)

4.2.2 Underlying principles

Command	Function / Explanation
Set bus address of mainboard	Defines the device address of *MembraSens* in a Modbus network. Then execute "Serial type 0" for Modbus RTU. Then set DIP switch 4 to ON and reboot to activate the selected bus address. (If the DIP switch 4 is off, then the address is 1.) See chapter 2.2.2.
Serial type	Sets the Modbus transmission mode. 0 (default value) corresponds to RTU, 1 for ASCII. The number of data bits must be 8, in both cases.
Operation mode	0: sleep mode, 2: standby mode, 3: work mode (default)

Command	Reg.-Adr.	W/R	Data Contents	AW	Data type	AR	A
Set bus address of mainboard	1110	W/R	1-247 (default: 31)	1	UNIT 16	1	0
Serial type	1111	W/R	0, 1	1	UNIT 16	1	0
Operation mode	1740	W/R	0, 2, 3	1	UNIT 16	1	1

4.2.3 Addressing the AFEs

The commands listed in the tables concerning an AFE are marked with "1" in the last column (A). The listed register address refers to the AFE 0. If one counts the number of registers AR, one would obtain the register address of the AFE 1, etc.

If several AFEs are connected, then it is always advisable to read the entire register area at once, respectively to write.

4.2.4 Reading out the sensor signals

Command	Function / Explanation
Get ppm	Displays the measured ppm value of the sensor on the corresponding AFE. The ppm value is calculated using the selected calc_x function, which can be set using the command "Calc_x for ppm calculation".
Get voltage	Displays the measured voltages after the transimpedance amplifier (TIA). These depend on the selected TIA-Gain. The values are in volts.
Get current	Displays the calculated sensor currents based on the measured voltages and the selected gain on the Transimpedance Amplifier (TIA). The values are in microamps.
Get current zero corrected	Displays the calculated sensor currents based on the measured voltages and TIA gain after subtracting the calculated sensor currents from the zero calibration (Calibrate zero point). The values are in microamps.
Get temperature	Displays the temperature measured on the AFE. The value is in ° C.

Command	Reg.-Adr.	W/R	Data Contents	AW	Data type	AR	A
Get ppm	1000	R	ppm value	1	FLOAT SGL	2	1
Get voltage	1010	R	Suspense Sens and Aux	2	FLOAT SGL	4	1
Get current	1030	R	Current Sens and Aux	2	FLOAT SGL	4	1
Get current zero corrected	1050	R	Current Sens and Aux	2	FLOAT SGL	4	1
Get temperature	1070	R	Temperature	1	FLOAT SGL	2	1

4.2.5 Configure the AFEs

Command	Function / Explanation
3E or 4E sensor type	The choice between 3-electrode and 4-electrode sensor can be set or read.
R_Load	The load resistance (R_Load) in the potentiostatic circuit can be set or read. The values 0-3 correspond to 10, 33, 50, 100 ohms.
TIA Gain (Sens + Aux)	The gain of the transimpedance amplifier (TIA) can be set or read. The values 0-7 correspond to the resistances: 350, undefined, 120, 35, 14, 7, 3.5, 2.75 kOhm.
TIA Gain (only Aux)	The gain of the transimpedance amplifier (TIA) for the aux signals alone can be set or read. The values 0-7 correspond to the resistances: 350, undefined, 120, 35, 14, 7, 3.5, 2.75 kOhm.
Internal zero	Writes or reads the internal zero point of the electronics: The values 0, 1, 2 correspond to 20%, 50%, 67% of V-Ref.
Bias voltage	Writes or reads the value of the bias voltage: 0 - 13 corresponds to: 0% - 24% of V_Ref.
Bias polarity	Writes or reads the bias polarity: 0 corresponds to a negative, 1 to a positive bias polarity. Only for sensors that require a bias voltage.
Calc_x for ppm calculation	The mathematical function for the ppm calculation can be selected or read out. The values 3-6 correspond to the functions Calc_3 - Cal_6.

Command	Reg.-Adr.	W/R	Data Contents	AW	Data type	AR	A
3E or 4E sensor type	1530	W/R	3, 4	1	UNIT 16	1	1
R_Load	1520	W/R	0 – 3	1	UNIT 16	1	1
TIA Gain (Sens + Aux)	1510	W/R	0 – 7	1	UNIT 16	1	1
TIA Gain (only Aux)	2020	W	0 – 7	1	UNIT 16	1	1
Internal zero	1710	W/R	0, 1, 2	1	UNIT 16	1	1
Bias voltage	1730	W/R	0 – 13	1	UNIT 16	1	1
Bias polarity	1720	W/R	0, 1	1	UNIT 16	1	1
Calc_x for ppm calculation	1540	W/R	3 – 6	1	UNIT 16	1	1

4.2.6 Configuration details

In the table bellow are the real values listed, which are represented by the data contents. The values in bold are the default values.

Command:	R_Load	TIA Gain	Internal zero	Bias voltage	Bias polarity	Calc_x
Unit:	**Ohm**	**kOhm**	**V**	**mV**	**-**	**-**
Value 0	10	350	**0.50 (20%)**	**0**	negative	
Value 1	**33**	NaN	1.25 (50%)	25	**positive**	
Value 2	50	120	1.68 (67%)	50		
Value 3	100	35		100		**Calc_3**
Value 4		**14**		150		Calc_4
Value 5		7		200		Calc_5
Value 6		3.5		250		Calc_6
Value 7		2.75		300		
Value 8				350		
Value 9				400		
Value 10				450		
Value 11				500		
Value 12				550		
Value 13				600		

Remarks

TIA Gain: If the maximal possible sensor current can exceed > 143 µA, then the resistance must be less than 14 kOhm, in the case of reducing gases (like CO). In the case of oxidizing gases (like NO2) is the limit at 96 µA.
Internal zero: A voltage range of 0 - 2.5 V is available. A sensor can generally swing to the positive or to the negative side, whereas one direction corresponds to a increasing gas concentration. With sensors for reducing gases (like CO) it is recommended to set the zero line at 20% of 2.5 V (thus 0.5 V), in the case of oxidizing gases (like NO2) at 67% of 2.5 V (thus 1.68 V).

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4.2.7 Save the configuration and temperature compensation

Command	Function / Explanation
Save configuration	Saves the active configuration and calibration to the EEPROM. This command is necessary if the configuration is changed without performing a calibration afterwards. The settings are saved in the user EEPROM and loaded after a restart.
Load configuration and calibration	Loads the last saved configuration and calibration from the user EEPROM. This command will be executed automatically after a reboot.
Load factory configuration and calibration	Loads the last saved configuration and calibration from the factory EEPROM.
Save factory configuration and calibration	Saves the active configuration and calibration to the factory EEPROM.

Command	Reg.-Adr.	W/R	Data Contents	AW	Data type	AR	A
Save configuration	1400	W	0	1	UNIT 16	1	1
Load configuration and calibration	1410	W	0	1	UNIT 16	1	1
Load configuration and calibration	1420	W	0	1	UNIT 16	1	1
Save factory configuration and calibration	2030	W	0	1	UNIT 16	1	1

4.2.8 Temperature compensation

Command	Function / Explanation
TCZ on/off	Activates or deactivates the temperature compensation of the baseline, or the current state can be read out.
TCS on/off	Activates or deactivates the temperature compensation of the sensitivity, respectively the current state can be read out.
TCG on/off	Activates or deactivates the temperature compensation of the gain of the H2 compensation, respectively the current state can be read out.
TCL on/off	Activates or deactivates the temperature compensation with L-type sensors, respectively the current state can be read out.
Coefficients for TCZ	Writes or reads the coefficients a1, a2, a3 (see chap. 5.7) for the temperature compensation curve of the baseline, which is a polynomial function of the 3. degree. a0 can not be set but is calculated so that the value of the compensation curve is 0 at the temperature at which the calibration occurred.
Coefficients for TCS	Writes or reads the coefficients a1, a2, a3 for the temperature compensation curve of the sensitivity, which is a polynomial function of the 3. degree. a0 can not be set but is calculated so that the value of the compensation curve is 0 at the temperature at which the calibration occurred.
Coefficients for TCG	Writes or reads the coefficients a1, a2, a3 for the temperature compensation curve of the gain in the H2 compensation, which is a polynomial function of the 3. degree. a0 can not be set but is calculated so that the value of the compensation curve is 0 at the temperature at which the calibration occurred.
Coefficients for TCL	Writes or reads of the baseline temperature compensation coefficient a1 for L-type sensors, which is a linear function. a0 can not be set but is calculated so that the value of the compensation function is 0 at the temperature at which the calibration occurred. 3 coefficients must be transferred.

Command	Reg.-Adr.	W/R	Data Contents	AW	Data type	AR	A
TCZ on/off	1550	W/R	0, 1	1	UNIT 16	1	1
TCS on/off	1560	W/R	0, 1	1	UNIT 16	1	1
TCG on/off	1570	W/R	0, 1	1	UNIT 16	1	1
TCL on/off	1580	W/R	0, 1	1	UNIT 16	1	1
Coefficients for TCZ	1590	W/R	a1, a2, a3	3	FLOAT SGL	6	1
Coefficients for TCS	1620	W/R	a1, a2, a3	3	FLOAT SGL	6	1
Coefficients for TCG	1650	W/R	a1, a2, a3	3	FLOAT SGL	6	1
Coefficients for TCL	1680	W/R	a1	3	FLOAT SGL	6	1

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4.2.9 Calibration with gas

Command	Function / Explanation
Start calibration mode	Starts the calibration mode.
End calibration mode	Ends the calibration mode. All calibrated values become active, the parameters are calculated and the calibration is saved in the user EEPROM. The calculated parameters can be read in registers starting from 3000.
Calibrate zero point	Captures the sensor signals at the zero point calibration (baseline of the sensor). With these values, the calculation of the ppm signal results in 0 ppm when no gas is present.
Calibrate point 1	Captures the sensor signals at the calibration point with target gas GA. These values are used to calculate the sensitivity of the sensor to the target gas to allow the ppm reading on exposure to the target gas.
Calibrate point 2	Captures the sensor signals at the calibration point with a gas mixture of GA and GB. These values are used to calculate the cross sensitivity of the sensor to the interfering gas GB to allow the ppm reading when exposed to a mixture of measurement and interference gas. This command is only needed for 4E sensors with hydrogen compensation.

Command	Reg.-Adr.	W/R	Data Contents	AW	Data type	AR	A
Start calibration mode	1200	W	0	1	UNIT 16	1	1
End calibration mode	1210	W	0	1	UNIT 16	1	1
Calibrate zero point	1220	W	0	1	UNIT 16	1	1
Calibrate point 1	1230	W/R	ppm for GA1	1	FLOAT SGL	2	1
Calibrate point 2	1240	W/R	ppm for GA2, ppm for GB2	2	FLOAT SGL	4	1

4.2.10 Example Calibration of the zero point and then with calibration gas

The following are the commands when calibrating an AFE with a 3-electrode sensor. The sensor is plugged into the AFE 0 and the calibration gas has a concentration of 102.3 ppm.

Condition at calibration point 0: Sensor is exposed to fresh ambient air or zero gas. Commands:

```
1200 write 0
1220 write 0
```

Condition at calibration point 1: Sensor is exposed to the calibration gas, which has a concentration of 102.3 ppm. Commands:

```
1230 write 102.3
1210 write 0
```

It is also possible to carry out the calibration with only one calibration point 1 alone and at any time.

4.2.11 Calibration values

This chapter lists the commands in order to read the calibration values. With these, the user can also specify (overwrite) the calibration values himself, including the sensor currents. This makes it possible to transfer calibration data from an external system to *MembraSens* without having to calibrate with gas. For this, as with a correct calibration, the calibration mode must be started and then terminated again.

Command	Function / Explanation
Start calibration mode	Starts the calibration mode. Only required if subsequently values will be written.
End calibration mode	Ends the calibration mode. All calibrated values become active, the parameters are calculated and the calibration is saved in the user EEPROM. The calculated parameters can be read in registers starting from 3000.
Value zero point calib	Values can be read to obtain the stored sensor signals of the zero point calibration. In calibration mode, values can be written to store any sensor currents as a zero point calibration.
Value point 1 calib	Values can be read to obtain the stored sensor signals at calibration point 1 with the target gas GA. Values can be written to store any sensor currents at calibration point 1 with target gas GA.
Value point 2 calib	Values can be read to obtain the stored sensor signals at calibration point 2. Values can be written to store any sensor currents at calibration point 2 with a mixture of GA and GB. This command is only needed for 4E sensors with hydrogen compensation.
Get calculated parameters	Reads the parameters calculated based on the calibration.

Command	Reg.-Adr.	W/R	Data Contents	AW	Data type	AR	A
Start calibration mode	1200	W	0	1	UNIT 16	1	1
End calibration mode	1210	W	0	1	UNIT 16	1	1
Value zero point calib	1300	W/R	µA for SEN, µA for AUX	2	FLOAT SGL	4	1
Value point 1 calib	1320	W/R	µA for SEN, µA for AUX, ppm for GA1	3	FLOAT SGL	6	1
Value point 2 calib	1350	W/R	µA for SEN, µA for AUX, ppm for GA2, ppm for GB2	4	FLOAT SGL	8	1
Get calculated parameters	3000	R	a, b, c, d, S	5	FLOAT SGL	10	1

To read calibration values, only execute the corresponding command without starting the calibration mode.
Example Reading the sensor currents and the gas concentration used for the AFE 0: `1320 read`

4.2.12 Alerting and EEPROM formatting

Command	Function / Explanation
Alarm on / off	Activates or deactivates the alarm, respectively the current state can be read out.
Alarm ppm level	Writes or reads the ppm limit from which an alarm is triggered.
Acknowledge alarm	Confirms an alarm. Otherwise, an alarm will be turned off after 5 minutes.
Format eeprom	Formats the EEPROM 0 or 1 with the value 0.

Command	Reg.-Adr.	W/R	Data Contents	AW	Data type	AR	A
Alarm on / off	1990	W/R	0, 1	1	UNIT 16	1	1
Alarm ppm level	2000	W/R	$0 – 3*10^{38}$	1	FLOAT SGL	2	1
Acknowledge alarm	1100	W	0	1	UNIT 16	1	0
Format eeprom	2050	W	0, 1	1	UNIT 16	1	1

4.2.13 Device data

Command	Function / Explanation
Serial Number	Serial number of the mainboard
Version SW	Version number of the firmware
Version HW	Hardware version
Production Code	Production code
Production date	Production date
Calibration date	To enter the calibration date
Part number	Part number
Runtime counter	To enter the operating hours
Code signature	To enter a signature
Sensor serial number	For entry of the serial number of the sensor
Sensor type	For entry of the sensor name
Upper measurement limit	For entry of the upper measuring limit of a sensor
Reload settings from flash	Load the values from 0 - 110 from flash memory to RAM
Save settings to flash	Stores the values from 0 - 110 in the flash memory

Command	Reg.-Adr.	W/R	Data Contents	AW	Data type	AR	A
Serial Number	0	W/R	Serial number	1	UNIT 32	2	0
Version SW	2	R	Software version	2	UNIT 16	2	0
Version HW	4	W/R	Hardware version	1	UNIT 32	2	0
Production Code	6	W/R	Production code	1	UNIT 32	2	0
Production date	8	W/R	Production date	1	UNIT 32	2	0
Calibration date	10	W/R	Calibration date	1	UNIT 32	2	0
Part number	12	W/R	Part number	1	UNIT 32	2	0
Runtime counter	14	W/R	Operating hours	1	UNIT 32	2	0
Code signature	16	W/R	Signature	1	UNIT 32	2	0
Sensor serial number	30	W/R	Sensor serial number AFE 0	1	UNIT 32	2	1
Sensor type	40	W/R	Sensor type AFE 0	1	String	8	1
Upper measurement limit	100	W/R	Upper measuring limit AFE 0	1	UNIT 32	2	1
Reload settings from flash	998	W		1	UNIT 16	1	0
Save settings to flash	999	W		1	UNIT 16	1	0

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5 ppm Calculations and Temperature Compensation

5.1 Calc_x: The Calculation Methods of the ppm-Concentration

With the function Calc_x for ppm calculation (register address 1540, see chap. 4.2.5)it can be determined how the gas concentration should be calculated from the raw signals. Which mathematical function one uses depends largely on the sensortype.

5.2 Overview

Function	Description
Calc_3	Calculation of the gas concentration with a 3-electrode sensor. Can also be used with 4-electrode sensors, which is then treated as a 3-electrode sensor.
Calc_4	Calculation of the gas concentration with a 4-electrode sensor with hydrogen compensation.
Calc_5	Calculation of the gas concentration with a 4-electrode sensor with temperature compensation.
Calc_6	Calculation of the gas concentration with a 4-electrode sensor, if only the auxiliary electrode is to be considered.

5.3 Calc_3

Formula	Explanation
$ppm=\left(\frac{I_s-I_{s,0}}{a}\right)\cdot TCS(T)+TCZ(T)$	Calculation of the gas concentration, if the value "3" was set with "Calc_x for ppm calculation".
$a=\frac{I_{s,1}-I_{s,0}}{GA_1}$	Calculation of the sensitivity. Will be performed at the end of the calibration.

Expression	Description	Source
I_s	Current at the sensing electrode (sensing)	Measuring-Signal
$I_{s,0}$	Baseline of the sensing electrode	Calibration zero point
$I_{s,1}$	Current at the sensing electrode during exposure with calibration gas	Calibration point 1
a	Sensitivity of the sensing electrode	Calibration point 1
GA_1	Concentration of the calibration gas A	Calibration point 1
$TCS(T)$	Temperature compensation of the sensitivity	
$TCZ(T)$	Temperature compensation of the baseline	

5.4 Calc_4

Formula	Explanation
$ppm=\left(\frac{(I_s-I_{s,0})-G\cdot(I_a-I_{a,0})\cdot TCG(T)}{S}\right)\cdot TCS(T)+TCZ(T)$	Calculation of the gas concentration, if the value "4" was set with "Calc_x for ppm calculation".
$a=\frac{I_{s,1}-I_{s,0}}{GA_1}$	Calculation of the sensitivity of the sensing electrode. Will be executed at the end of the calibration.
$c=\frac{I_{a,1}-I_{a,0}}{GA_1}$	Calculation of the sensitivity of the auxiliary electrode. Will be executed at the end of the calibration.
$b=\frac{(I_{s,2}-I_{s,0})-a\cdot GA_2}{GB_2}$	Calculation of the cross-sensitivity of the sensing electrode. Will be executed at the end of the calibration.
$d=\frac{(I_{a,2}-I_{a,0})-c\cdot GA_2}{GB_2}$	Calculation of the cross-sensitivity of the auxiliary electrode. Will be executed at the end of the calibration.
$G=\frac{b}{d}$	Gain of the cross-sensitivity compensation.
$S=(a-G\cdot c)$	Calculation of the sensor sensitivity. Will be executed at the end of the calibration.

Expression	Description	Source
I_s	Current at the sensing electrode (sensing)	Measuring-Signal
I_a	Current at the auxiliary electrode (Auxiliary)	Measuring-Signal
$I_{s,0}$, $I_{a,0}$	Baseline of the sensing and auxiliary electrode	Calibration zero point
$I_{s,1}$, $I_{a,1}$	Currents at the sensing and auxiliary electrode during exposure with calibration gas	Calibration point 1
a , c	Sensitivity of the sensing and auxiliary electrode	Calibration point 1
GA_1	Concentration of the calibration gas A	Calibration point 1
$I_{s,2}$, $I_{a,2}$	Currents at the sensing and auxiliary electrode during exposure with calibration gas	Calibration point 2
GA_2 , GB_2	Concentrations of calibration gases A and B	Calibration point 2
$TCG(T)$	Temperature compensation of the gain	
$TCS(T)$	Temperature compensation of the sensitivity	
$TCZ(T)$	Temperature compensation of the baseline	

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5.5 Calc_5

Formula	Explanation
$ppm=\left(\frac{I_s-TCL(T)\cdot I_a-\Delta I_0}{a}\right)\cdot TCS(T)$	Calculation of the gas concentration, if the value "5" was set with "Calc_x for ppm calculation".
$\Delta I_0=I_{s,0}-I_{a,0}$	Calculation of the baseline-offset. Will be executed at the end of the calibration.
$a=\frac{I_{s,1}-I_{a,1}-\Delta I_0}{GA_1}$	Calculation of the sensitivity of the sensing electrode. Will be executed at the end of the calibration.

Expression	Description	Source
I_s	Current at the sensing electrode (sensing)	Measuring-Signal
I_a	Current at the auxiliary electrode (Auxiliary)	Measuring-Signal
$I_{s,0}$, $I_{a,0}$	Baseline of the sensing and auxiliary electrode	Calibration zero point
ΔI_0	Baseline-Offset	Calibration zero point
$I_{s,1}$, $I_{a,1}$	Currents at the sensing and auxiliary electrode during exposure with calibration gas	Calibration point 1
a	Sensitivity of the sensing electrode	Calibration point 1
GA_1	Concentration of the calibration gas	Calibration point 1
$TCL(T)$	Temperature compensation of the baseline	
$TCS(T)$	Temperature compensation of the sensitivity	

5.6 Calc_6

Formula	Explanation
$ppm=\left(\frac{I_a-I_{a,0}}{c}\right)\cdot TCS(T)+TCZ(T)$	Calculation of the gas concentration, if the value "6" was set with "Calc_x for ppm calculation".
$c=\frac{I_{a,1}-I_{a,0}}{GA_1}$	Calculation of the sensitivity of the auxiliary electrode. Will be executed at the end of the calibration.

Expression	Description	Source
I_a	Current at the auxiliary electrode (Auxiliary)	Measuring-Signal
$I_{a,0}$	Base line of the auxiliary electrode	Calibration zero point
$I_{a,1}$	Current at the auxiliary electrode during exposure with calibration gas	Calibration point 1
c	Sensitivity of the auxiliary electrode	Calibration point 1
GA_1	Concentration of the calibration gas	Calibration point 1
$TCS(T)$	Temperature compensation of the sensitivity	
$TCZ(T)$	Temperature compensation of the baseline	

5.7 Temperature Compensation: The Models

Temperature-dependent parameters such as sensitivity, baseline and gain can be easily compensated using a polynomial of the 3. degree:

$$TCx(T)=k_0+k_1\cdot T+k_2\cdot T^2+k_3\cdot T^3$$

The coefficients k_1, k_2 formula and k_3 determined by the user. The first coefficient k_0 is automatically calculated so that no temperature compensation contribution is made at the calibration temperature.

The methodology for temperature compensation is described in detail in the document *Application_Note_MEM7* .

The temperature compensation can be switched on and off (see chap. 5.2.7) The ppm calculation also works when the temperature compensation is switched off.

6 Appendix Default Values of the AFE Configuration

An AFE is configured as follows at MEMBRAPOR if it is not explicitly configured for a sensor:

```
* Standard configuration for reducing gases (like CO, H2S...)

3E or 4E sensor type
Write  4
* 4-Electrode-Sensor

R_Load
Write  0
* 10 Ohm load resistor

TIA Gain (Sens+Aux)
Write  4
* 14 kOhm for currents up to 178 uA

Internal zero
Write  0
* Internal zero at 0.5 V

Bias voltage
Write  0
* No bias voltage

Bias polarity
Write  1
* If once a bias would be set it would be most probably positive

Calc_x for ppm calculation
Write  4
* The formula for 4-electrode sensors with H2-compensation

Value zero point calib
Write  0      0

Value point 1 calib
Write  0      0      0

Value point 2 calib
Write  0      0      0      0

TCZ on/off
Write  0
```

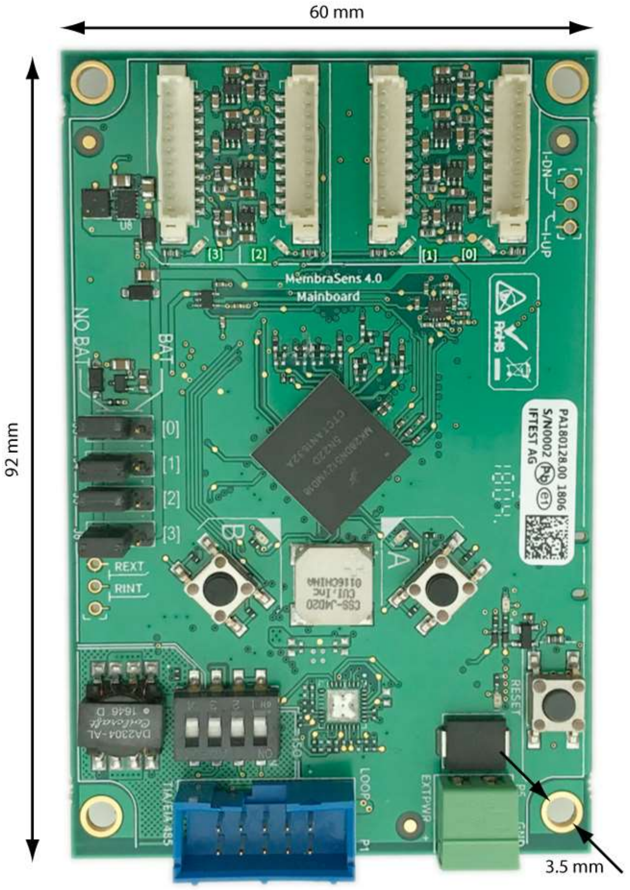
```
TCS on/off
Write  0

TCG on/off
Write  0

TCL on/off
Write  0


Coefficients for TCZ
Write  -0.006163688164     -0.001726806527     -0.000040530951

Coefficients for TCS
Write  -0.016721323021     0.000341633548      -0.000002562642

Coefficients for TCG
Write  -0.003361023384     -0.000061501354     -0.000000537826

Coefficients for TCL
Write  0.010

Alarm on/off
Write  0.000

Alarm ppm level
Write  1000000

Save configuration
Write  1

Save factory configuration and calibration
Write  1

End

--------------------------------------
```

```
* Configuration of AFEs with sensors for oxidizing gases (like NO2).


3E or 4E sensor type    4
Write 3
* 3-Elektroden-Sensor

R_Load      4
Write 0
* 10 Ohm Lastwiderstand

TIA Gain (Sens+Aux)     4
Write 4
* 14 kOhm für Ströme bis 143 uA

Internal zero     4
Write 2
* Internes Null bei 1.675 V (Fraction 0.67 of V_Ref)

Bias voltage      4
Write 0
* Keine Bias-Spannung
```

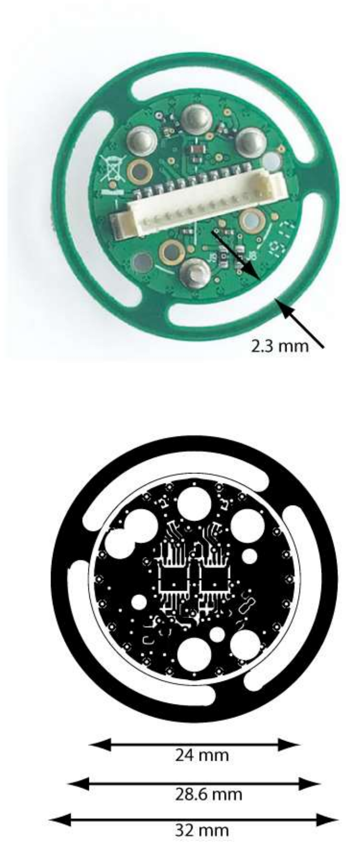
```
Bias polarity     4
Write 1
* Falls einmal Bias gesetzt wird, dann wird er meist positiv sein


Calc_x for ppm calculation    4
Write 3
* Die Formel für 3-Elektroden-Sensor

Value zero point calib  4
Write 0     0

Value point 1 calib     4
Write 0     0     0

Value point 2 calib     4
Write 0     0     0     0

TCZ on/off  4
Write 0

TCS on/off  4
Write 0

TCG on/off  4
Write 0

TCL on/off  4
Write 0

Coefficients for TCZ    4
Write -0.006163688164   -0.001726806527   -0.000040530951

Coefficients for TCS    4
Write -0.016721323021   0.000341633548    -0.000002562642

Coefficients for TCG    4
Write -0.003361023384   -0.000061501354   -0.000000537826

Coefficients for TCL    4
Write 0.010

Alarm on/off      4
Write 0.000

Alarm ppm level   4
Write 1000000

Save configuration      4
Write 1

Save factory configuration and calibration      4
Write 1

End
```

7 Annex of Technical Drawings

Mainboard



AFE



8 Additional Information

8.1 Related Documents

- *MembraSens* Test Panel

8.2 Contact Information

For more information contact our technical support at MEMBRAPOR.

- www.membrapor.ch/technical-support

9 History of Changes

Revision date:	Revision Doc	Version hardware	Changes
20.07.2018	07/2018	4.0	Provisonal Release of a Draft
13.09.2018	09/2018	4.0	Correction MODBUS interface.
21.01.2019	01/2019	4.0	Chapter 9
21.02.2019	02/2019	4.0	Correction Tab.6. Supplements in Chap. 4.
19.07.2019	07/2019	4.0	Chapter 3.1 reworded. Technical drawings added. Official release of version 4.
30.08.2021	08/2021	4.0	Chapter 4.2.6 introduced. Correction default values of dip-switch (2.2.3). Adding configuration in chapter 6.
24.01.2024	01/2024	4.0	Default value in register 1110
11.04.2025	04/2025	4.0	Error in Calc_4 formula corrected (subtraction of the baseline of the auxiliary electrode). Chapter 4.2.6. “Configuration details“ added.

Tab 6: Revision history

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April 2025

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